Life Insurance foundamentals

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The biometric model

- The **biometric model** focuses on the modelization of the random variable X that measures the death age of a person.
- X is defined in the set of positive real numbers, although in practice it is accepted that there is a limit in the death age, called **final age** ω, which is the maximum age that a person will reach. Therefore, X ranges between $(0, \omega)$.
- We will also consider the **residual life** at age x, called $T(x)$, which is the remaining lifetime that a person aged x will survive. Specifically,

$$T(x) = X - x$$

which takes values in the interval $(0, \omega - x)$.

The biometric model: hypothesis

- **Homogeneity**: the individuals in the portfolio are homogeneous, this means that the statistical behaviour of their death ages is the same, i.e. if x_i and x_j are the death ages of two individuals i and j, then $F_{x_i}(x) = F_{x_j}(x)\ \forall\ x \in \mathbb{R}^+$, where F is the distribution function of the death age.
- **Independency**: the variables that describe the death age of two individuals are statistically independent. If x_i and x_j are the death ages of two individuals i and j, then $F_{x_i|x_j=y}(x) = F(x)\ \forall\ x, y \in \mathbb{R}^+$.
- **Stationarity**: the biometric properties of the individuals do not depend on their birth date, but only on their age. This hypothesis is accepted for short time periods.

Biometric functions

- Let us asume that F is the **distribution function of the death age** x,

$$F(x) = P(X \leq x)$$

i.e. the probability that the death age X will not be greater than x.

- On the other hand, the **survival function** $S(x)$ provides the probability that the death age will be lower than x

$$S(x) = P(X > x) = 1 - F(x)$$

where $S(0) = 1$ and $S(\omega) = 0$.

Biometric functions

- We are also interested in the distribution function of the **residual life** at x, $F_{T(x)}$, that will be denoted as G_x, specifically

$$G_x(t) = P(T(x) \leq t) = P(X - x \leq t | X > x) =$$

$$= P(X \leq x + t | X > x) = \frac{F(x+t) - F(x)}{1 - F(x)}$$

for $0 < t < \omega - x$.

Temporal probabilities

- **Temporal probability of death** at age x: ${}_nq_x$ is the probability that an individual who has survived until age x will die before age $x + n$. Therefore, it is a conditional probability of death.

$$ {}_nq_x = P(x < X \leq x + n | X > x) = \frac{F(x+n) - F(x)}{1 - F(x)} $$

- **Temporal probability of survival** at age x: ${}_np_x$ is the probability that an individual who has survived until age x will reach age $x + n$. Therefore, it is a conditional probability of survival.

$$ {}_np_x = P(X > x + n | X > x) = \frac{1 - F(x+n)}{1 - F(x)} = 1 - {}_nq_x $$

Temporal probabilities

- Although temporal probabilities are defined for a period of n years, in practical applications it is very frequent to work with **annual probabilities**. In that case, we omit the subindex that corresponds to the time period, specificaly, we write p_x and q_x (instead of ${}_1p_x$ and ${}_1p_x$).
- It can be proved that

$$
\begin{aligned}
{}_np_x &= P(X > x + n | X > x) = \frac{P(X>x+n)}{P(X>x)} = \\
&= \frac{P(X>x+n)}{P(X>x+n-1)} \cdot \frac{P(X>x+n-1)}{P(X>x+n-2)} \cdot \cdots \frac{P(X>x+1)}{P(X>x)} = \\
&= p_{x+n-1} \cdot p_{x+n-2} \cdot \cdots \cdot p_x = \prod_{t=0}^{n-1} p_{x+t}
\end{aligned}
$$

Temporal probabilities

- Based on that result, we say that the temporal probability of survival fullfils the property of **separability**

$$ {}_n p_x = {}_k\, p_x \cdot {}_{n-k}\, p_{x+k} $$

Proof:

$$ {}_n p_x = \frac{P(X > x + n)}{P(X > x)} = \frac{P(X > x + k)}{P(X > x)} \cdot \frac{P(X > x + n)}{P(X > x + k)} = $$

$$ = {}_k\, p_x \cdot {}_{n-k}\, p_{x+k} $$

Temporal probabilities

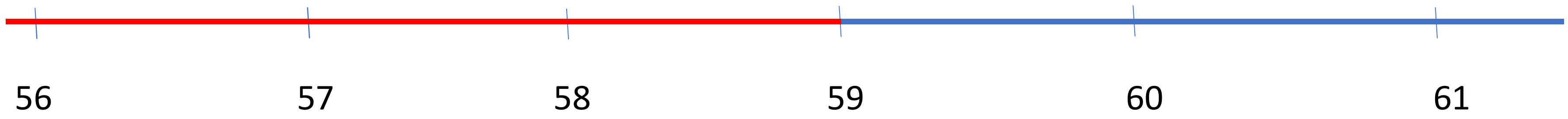
- Similarly, for the temporal probability of death we have

$$ {}_nq_x = \frac{P(x < X \leq x+n)}{P(X > x)} = \frac{P(x < X \leq x+k) + P(x+k < X \leq x+n)}{P(X > x)} $$

$$ = \ {}_kq_x + {}_kp_x \cdot {}_{n-k}q_{x+k} $$

Therefore, we say that **it does not fulfill the separability property**.

Example



$${}_5p_{56} = {}_3p_{56} \cdot {}_2p_{59}$$

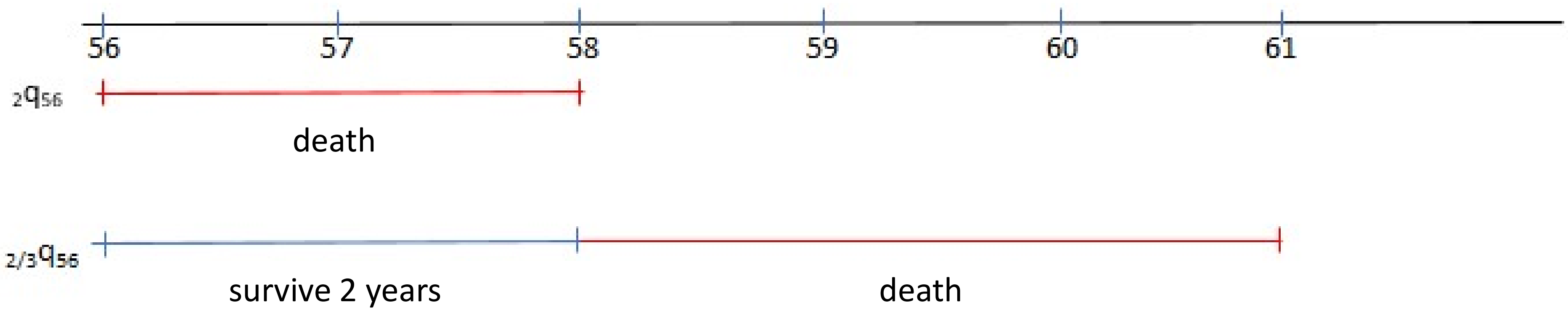
$${}_5q_{56} = {}_3q_{56} + {}_3p_{56} \cdot {}_2q_{59}$$

Temporal and deferred probabilities

- The **temporal and deferred probabilities** of death consider the posibility that an individual will survive during a period of time and die afterwards.
- **Temporal and deferred probability of death for an insured aged x, denoted** as ${}_{m/n}q_x$: is the probability that an insured aged x will die between age $x+m$ and $x+m+n$. It is defined as

$$
{}_{m/n}q_x = \frac{P(x+m < X \leq x+m+n)}{P(X > x)} = \frac{F(x+m+n) - F(x+m)}{1 - F(x)}
$$

Example



Temporal and deferred probabilities

- Temporal and **deferred probabilities** can be expressed **as a function of temporal probabilities**. Specifically we have that

- $_{m/n}q_x = {}_m p_x \cdot {}_n q_{x+m}$
- $_{m/n}q_x = {}_{m+n} q_x - {}_m q_x$
- $_{m/n}q_x = {}_m p_x - {}_{m+n} p_x$

Temporal and deferred probabilities

- $_{m/n}q_x = {}_m p_x \cdot {}_n p_{x+m}$

$$_{m/n}q_x = \frac{P(x+m < X \leq x+m+n)}{P(X > x)}$$

$$= \frac{P(X > x+m)}{P(X > x)} \cdot \frac{P(x+m < X \leq x+m+n)}{P(X > x+m)} = {}_m p_x \cdot {}_n q_{x+m}$$

Temporal and deferred probabilities

- $_{m/n}q_x = {}_{m+n}\, q_x - {}_m q_x$

$$_{m/n}q_x = \frac{P(x + m < X \leq x + m + n)}{P(X > x)}$$

$$= \frac{P(x < X \leq x + m + n)}{P(X > x)} - \frac{P(x < X \leq x + m)}{P(X > x)} = {}_{m+n}\, q_x - {}_m q_x$$

And therefore… $_{m/n}q_x = {}_m\, p_x - {}_{m+n}p_x$

Temporal and deferred probabilities

- The **deferred probability of death** for an insured aged x, denoted as ${}_{m/}q_x$, is the probability that the insured will survive until $x + m$, and die during the next year (n is assumed to be equal to 1).

$$ {}_{m/}q_x = \frac{P(x + m < X \leq x + m + 1)}{P(X > x)} $$

- So, we have
- ${}_{m/}q_x = {}_{m}\,p_x \cdot q_{x+m}$
- ${}_{m/}q_x = {}_{m+1}\,q_x - {}_{m}q_x$
- ${}_{m/}q_x = {}_{m}\,p_x - {}_{m+1}p_x$

Temporal and deferred probabilities

- The **temporal probability** of death can be written **as a function of deferred probabilities**, specifically

$$ {}_{n}q_{x} = \sum_{t=0}^{n-1} {}_{t/}q_{x} $$

Temporal and deferred probabilities

- ${}_nq_x = \sum_{t=0}^{n-1} {}_{t/}\, q_x$

$$
\begin{aligned}
{}_nq_x &= \frac{P(x < X \leq x+n)}{P(X > x)} \\
&= \frac{P(x < X \leq x+1) + \ldots + P(x+n-1 < X \leq x+n)}{P(X > x)} \\
&= \frac{P(x < X \leq x+1)}{P(X > x)} + \ldots + \frac{P(x+n-1 < X \leq x+n)}{P(X > x)} \\
&= {}_{0/}\, q_x + {}_{1/}\, q_x + \ldots + {}_{n-1/}\, q_x = \sum_{t=0}^{n-1} {}_{t/}\, q_x .
\end{aligned}
$$

Force of mortality

- The **force of mortality**, μ_x or $\mu(x)$, is the instantaneous rate of mortality at a certain age.

$$\mu(x) = \lim_{\Delta t \to 0} \frac{{}_{\Delta t}q_x}{\Delta t}$$

It can be proved that

$${}_{\Delta t}q_x = P(X \leq x + \Delta t | X > x) = \frac{F(x + \Delta t) - F(x)}{1 - F(x)}$$

$$\frac{{}_{\Delta t}q_x}{\Delta t} = \frac{1}{1 - F(x)} \cdot \frac{F(x + \Delta t) - F(x)}{\Delta t}$$

Force of mortality

- If we calculate the limit, we have that $\mu(x) = \frac{f(x)}{1-F(x)}$
- Based on that result, it can be proved that

$$S(x) = 1 - F(x) = e^{-\int_0^x \mu(y)dy}$$

$$\mu(x) = -\frac{d}{dx}\ln S(x)$$

$$_{n}p_x = e^{-\int_x^{x+n} \mu(y)dy}$$

$$g_x(t) = {}_t\, p_x \cdot \mu(x+t)$$

where $g_x(t)$ is the density function of the residual life $T(x)$.

Force of mortality

- Proof

$$\int_0^x \mu(y)dy = \int_0^x \frac{f(y)}{1-F(y)}dy = -\ln[1-F(y)]\Big|_0^x =$$

$$= -\ln[1-F(x)]$$

$$\exp(\ln[1-F(x)]) = exp\left(-\int_0^x \mu(y)dy\right)$$

$$S(x) = 1-F(x) = e^{-\int_0^x \mu(y)dy}$$

Force of mortality

- Proof

$$\mu(x) = \frac{f(x)}{1 - F(x)} = -\frac{S'(x)}{S(x)} = -\frac{d}{dx} \ln S(x)$$

Force of mortality

- Proof

$$ {}_{n}p_{x} = \frac{1 - F(x+n)}{1 - F(x)} = \frac{e^{-\int_0^{x+n} \mu(y)dy}}{e^{-\int_0^{x} \mu(y)dy}} = $$

$$ = e^{-\int_0^{x+n} \mu(y)dy + \int_0^{x} \mu(y)dy} = $$

$$ = e^{-\int_x^{x+n} \mu(y)dy} $$

Force of mortality

- Proof: we recall that $g_x(t)$ is the density function of the residual life $T(x)$. We saw that the distribution function, $G_x(t)$ is equal to

$$G_x(t) = \frac{F(x+t) - F(x)}{1 - F(x)}$$

For $0 < t < \omega - x$. Therefore,

$$g_x(t) = \frac{f(x+t)}{1-F(x)} = \frac{1-F(x+t)}{1-F(x)} \cdot \frac{f(x+t)}{1-F(x+t)} = {}_t\, p_x \cdot \mu(x+t)$$

Life expectancy

- When a person decides to underwrite a life Insurance, one of the most important factors in order to calculate the premium is **the age of the policyholder**. Based on that, the **life expectancy** can be calculated.
- We recall that the distribution funciton of the residual life at age x, $G_x(t)$ equals

$$G_x(t) = P(T(x) \leq t) = P(X - x \leq t | X > x) = \frac{F(x+t) - F(x)}{1 - F(x)}$$

- Therefore, we have that $G_x(t) = {}_t\,q_x$

Life expectancy

- By calculating the first derivative, we have

$$g_x(t) = \frac{f(x+t)}{1-F(x)} = {}_t\,p_x \cdot \mu(x+t)$$

- The **life expectancy at age *x*** is the expected value of *T(x)*

$$\bar{e}_x = E[T(x)] = \int_0^{\omega - x} t \cdot g_x\,(t)dt = \int_0^{\omega - x} t \cdot {}_t\,p_x \cdot \mu(x+t)\,dt$$

Life expectancy

- And the **variance** of the residual life is

$$E[T^2(x)] = \int_0^{\omega - x} t^2 \cdot g_x(t) dt = \int_0^{\omega - x} t^2 \cdot {}_t p_x \cdot \mu(x+t)\, dt =$$

$$= -[t^2 \cdot {}_t p_x]_0^{\omega - x} + 2 \int_0^{\omega - x} t \cdot {}_t p_x \, dt = 2 \int_0^{\omega - x} t \cdot {}_t p_x \, dt$$

therefore

$$Var[T(x)] = 2 \int_0^{\omega - x} t \cdot {}_t p_x \, dt - \bar{e}_x^2$$

Life tables

- A **life table** is a table that shows, for each age, the probability that a person of that age will die before their next birthday, as well as additional other biometric functions (such as the survival probability, life expectancy, etc).
- One of these biometric functions is the **cohort function** l(x) which is the number of people who survived until age x from a cohort of individuals who are born at the same time (l(0) is the initial size of the cohort).
- We also have that $l(\omega) = 0$ and $l(x) = l(0) \cdot S(x)$.
- Usually, it is assumed that $l(0) = 100000$.

Life tables

- The function ${}_n d_x$ is the **number of deaths** between age x and x+n. Specifically, we have that ${}_n d_x = l(x) - l(x+n)$. When $n = 1$, we write directly d_x.

- Similarly, we have that

$$ {}_n p_x = \frac{l(x+n)}{l(x)} $$

$$ {}_n q_x = \frac{l(x) - l(x+n)}{l(x)} = \frac{{}_n d_x}{l(x)} $$

- Moreover, as we know $\mu(x) = -\frac{S'(x)}{S(x)}$. Therefore, $\mu(x) = -\frac{l'(x)}{l(x)}$